# Spatial Analysis

September 10, 2023

## 1 Introduction

We compare the ulcerative and intact regions of a piece of tissue as follows. First, we run a custom pipeline (https://github.com/youyifong/tsp) on an image to produce a database of cells in the tissue for each marker of interest. The pipeline starts with image registration, which transforms different marker images into one coordinate system so that we can compare the distribution of different markers from the same piece of tissue. After image registration, we perform cell segmenation using a Cellpose model fine-tuned with a set of expert-annotated training images of immune cells in skin tissues. For each cell thus identified, we compute the distance from its center to the skin boundary and the DEJ line, whether it is located in the dermis or epidermis and whether it is in the ulcerative or intact region, and the intensity of staining. Given the database, we run statistical analyses in R (https://cran.r-project.org/) to compare the distributions of immune cells between the ulcerative and intact regions tabularly and graphically. Graphically, we represent the distribution of a type of immune cells using smoothed scaled histogram (https://github.com/youyifong/kyotil), in which the y-axis is the density of immune cells, computed as the number of cells per 100x100 um2 tissue, and the x-axis is the depth of tissue in100um intervals, either with reference to the skin boundary or the DEJ line.

## 2 Methods

Issues

- Need to review length
  - For LS, epi is a lot wider than dermis. How do we draw the length?
  - DD intact, length correct? CD3 peak seems too high in Fig 1
- For JM, region 2 is excluded from analysis.
- Need to review segmentation. flow .5 seems to results in many more cells than flow .4, e.g. for CD11c in DD

For each sample, we will compare the distribution of each markers in the normal and lesion regions.

- Markers for DD: CD3, CD4, CD8a, CD11c, CD15, CD45, CD68, CD69, CD123, CD279
- Markers for CF: CD1a, CD3, CD4, CD8a, CD11c, CD15, CD45, CD69, CD123, CD279 (CD68 did not pass QC)

For CD123, segmentation is done at flow=0.5 to get more masks in the bright area and a threshold of 5000 is set for total intensity to remove masks in the debris field in the lesion area.

The size of the skin tissue lesion area is determined by morphology using wheat germ agglutinin (WGA) flourescent stain. ImageJ is used to demarcate the lesion area, then a straight line ROI is drawn to span

from one side of the lesion to the other side of the lesion, with the endpoints of the line positioned at the outermost skin boundary.

Some considerations on region boundary and distance metric:

1. Should we draw the boundary between normal and lesion based on WGA only or all markers?
2. Shall each ROI be convex?

For (1), the two choices lead are answering different questions. Drawing boundary based on WGA only answers a stronger question than drawing the boundary based on all markers.

For (2), a convex shape is a shape that a line connecting any two points in the shape completely lies within the shape. Practically speaking, in this context it is equivalent to drawing a straight line to separate normal from lesion. Pros for straight boundary: the span between any cell and the closest point on the skin lies completely within the region. So the distance measure is well defined. In contrast, if a line from a cell to its closest point on the skin surface cuts through both normal and lesion regions, the interpretation of distance to skin surface is muddled. Pros for curved boundary: Regions (which is a type of neighborhoods) do not necessarily have geometric shapes.

These two questions are also intertwined. If we use WGA only to draw boundaries, we can only use straight line boundaries because the difference between normal and lesion is only visible in epidermis in WGA staining. A straight line can be drawn in epidermis and extend to dermis, but not a curve.

# 3 Results

| | Counts | | | | Counts/mm2 | | | |
|---|---|---|---|---|---|---|---|---|
| | normal | | lesion | | normal | | lesion | |
| | epi | dermis | epi | dermis | epi | dermis | epi | dermis |
| CD11c | 333 | 2295 | 8194 | 20682 | 2841 | 5398 | 6301 | 5364 |
| CD123 | 131 | 879 | 1350 | 8038 | 1117 | 2067 | 1038 | 2084 |
| CD15 | 6 | 147 | 4449 | 1918 | 51 | 345 | 3421 | 497 |
| CD279 | 9 | 346 | 68 | 2168 | 76 | 813 | 52 | 562 |
| CD3 | 59 | 1134 | 92 | 4720 | 503 | 2667 | 70 | 1224 |
| CD4 | 140 | 1721 | 245 | 9286 | 1194 | 4048 | 188 | 2408 |
| CD45 | 341 | 2540 | 6082 | 15965 | 2909 | 5975 | 4677 | 4140 |
| CD68 | 112 | 619 | 1305 | 7910 | 955 | 1456 | 1003 | 2051 |
| CD69 | 17 | 245 | 27 | 2100 | 145 | 576 | 20 | 544 |
| CD8a | 35 | 639 | 67 | 3394 | 298 | 1503 | 51 | 880 |
| area | 0.12 | 0.43 | 1.30 | 3.86 | | | | |

Table 1: DD

| | Counts | | | | Counts/mm2 | | | |
|---|---|---|---|---|---|---|---|---|
| | normal | | lesion | | normal | | lesion | |
| | epi | dermis | epi | dermis | epi | dermis | epi | dermis |
| CD11c | 597 | 8839 | 1412 | 4600 | 1380 | 4690 | 3494 | 4555 |
| CD123 | 76 | 3680 | 324 | 1298 | 175 | 1952 | 801 | 1285 |
| CD15 | 4 | 1614 | 556 | 302 | 9 | 856 | 1376 | 299 |
| CD1a | 70 | 28 | 11 | 48 | 161 | 14 | 27 | 47 |
| CD279 | 110 | 1305 | 46 | 852 | 254 | 692 | 113 | 843 |
| CD3 | 381 | 7847 | 290 | 5424 | 881 | 4163 | 717 | 5371 |
| CD4 | 351 | 6876 | 281 | 3999 | 811 | 3648 | 695 | 3960 |
| CD45 | 536 | 9636 | 894 | 6015 | 1239 | 5113 | 2212 | 5956 |
| CD8a | 213 | 4280 | 191 | 3287 | 492 | 2271 | 472 | 3255 |
| area | 0.43 | 1.88 | 0.40 | 1.01 | | | | |

Table 2: CF

| | DD | CF | JM | CF/DD | JM/DD |
|---|---|---|---|---|---|
| tissue width | 2861 | 1202 | 1154 | 0.42 | 0.40 |
| CD3 to CD15 (dist2skin) | 311 | 211 | 529 | 0.68 | 1.70 |
| CD3 to CD15 (dist2DEJ) | 207 | 166 | 477 | 0.80 | 2.30 |
| CD3 epi | 70 | 717 | 241 | 10.24 | 3.44 |
| CD3 dermis | 1224 | 5371 | 2561 | 4.39 | 2.09 |
| CD3 e/d | 1.17 | 0.77 | 0.62 | 0.66 | 0.53 |
| CD15 epi | 3421 | 1376 | 1393 | 0.40 | 0.41 |
| CD15 dermis | 497 | 299 | 270 | 0.60 | 0.54 |
| CD15 e/d | 0.50 | 0.62 | 0.42 | 1.24 | 0.84 |
| CD123 epi | 1038 | 801 | 1257 | 0.77 | 1.21 |
| CD123 dermis | 2084 | 1285 | 3013 | 0.62 | 1.45 |
| CD123 e/d | 6.88 | 4.60 | 5.16 | 0.67 | 0.75 |
| CD11c epi | 6301 | 3494 | 2253 | 0.55 | 0.36 |
| CD11c dermis | 5364 | 4555 | 3651 | 0.85 | 0.68 |
| CD11c e/d | 0.09 | 0.57 | 0.08 | 6.33 | 0.89 |

Table 3: Ulcerative. The cell counts are per mm2. The widths and seperation are measured in um.

| | DD | CF | JM | CF/DD | JM/DD |
|---|---|---|---|---|---|
| tissue width | 251 | 1531 | 2114 | 6.10 | 8.42 |
| CD3 to CD15 (dist2skin) | 51 | -82 | 140 | -1.61 | 2.75 |
| CD3 to CD15 (dist2DEJ) | 52 | -22 | 23 | -0.42 | 0.44 |
| CD3 epi | 503 | 881 | 675 | 1.75 | 1.34 |
| CD3 dermis | 2667 | 4163 | 3883 | 1.56 | 1.46 |
| CD3 e/d | 0.53 | 0.29 | 0.57 | 0.55 | 1.08 |
| CD15 epi | 51 | 9 | 16 | 0.18 | 0.31 |
| CD15 dermis | 345 | 856 | 88 | 2.48 | 0.26 |
| CD15 e/d | 0.54 | 0.09 | 0.43 | 0.17 | 0.80 |
| CD123 epi | 1117 | 175 | 749 | 0.16 | 0.67 |
| CD123 dermis | 2067 | 1952 | 1743 | 0.94 | 0.84 |
| CD123 e/d | 0.15 | 0.01 | 0.18 | 0.07 | 1.20 |
| CD11c epi | 2841 | 1380 | 1666 | 0.49 | 0.59 |
| CD11c dermis | 5398 | 4690 | 2945 | 0.87 | 0.55 |
| CD11c e/d | 0.09 | 11.50 | 0.30 | 127.78 | 3.33 |

Table 4: Intact. The cell counts are per mm2. The widths and seperation are measured in um.

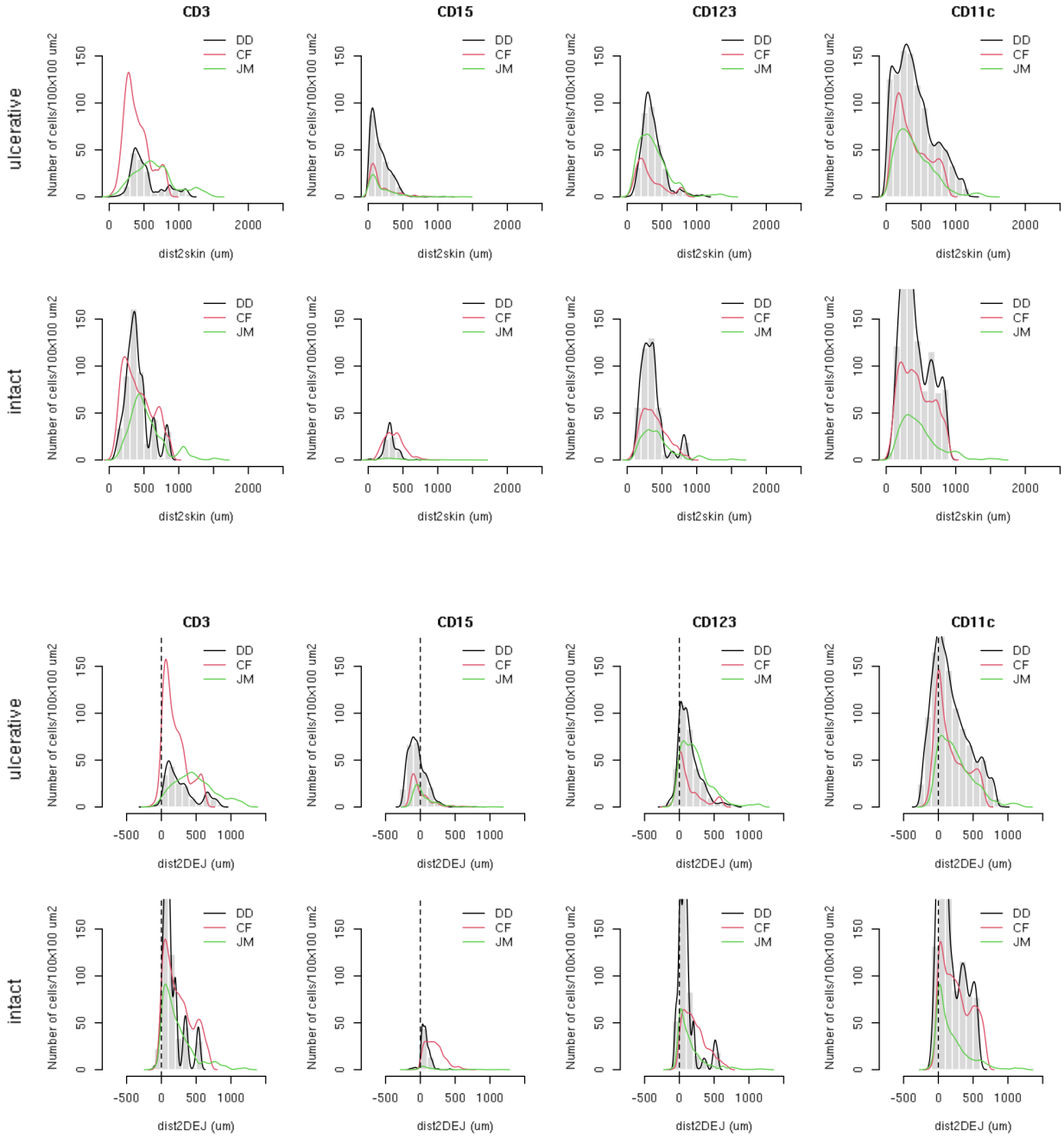



Figure 1: Number of cells per 100x100 um2 tissue as a function of distance to DEJ or skin. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width. For example, CD123 epi count 1038/2862*100=36.

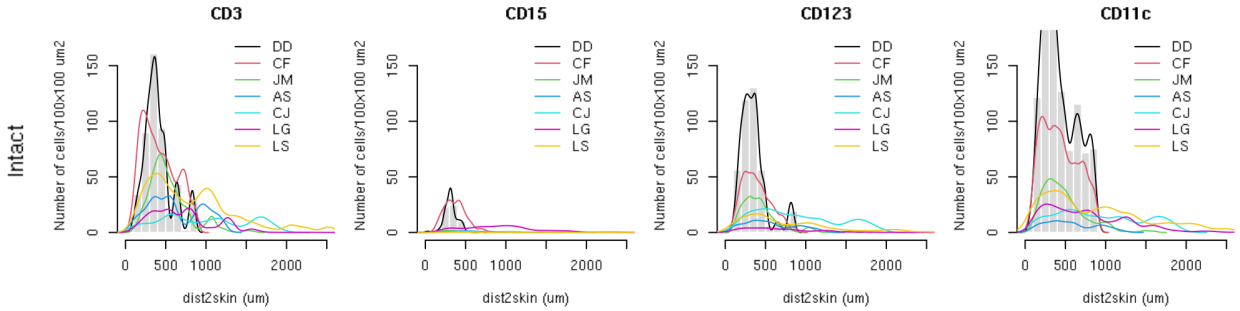



Figure 2: Number of cells per 100x100 um2 tissue as a function of distance to DEJ or skin. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width.

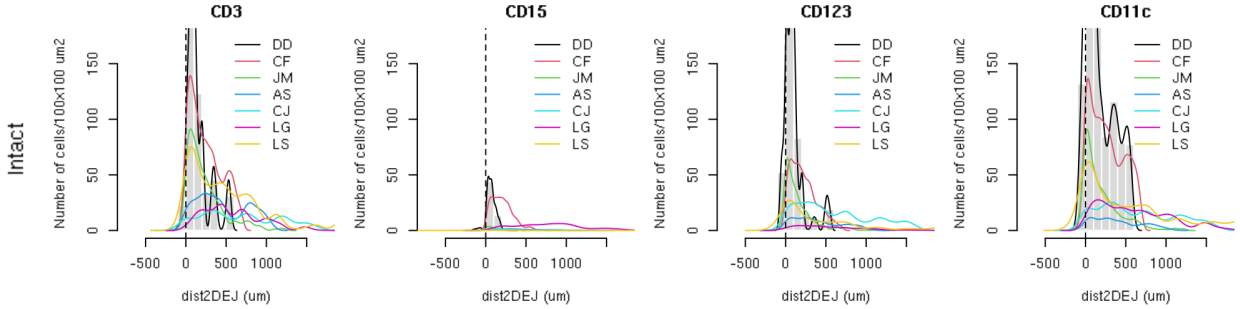



Figure 3: Number of cells per 100x100 um2 tissue as a function of distance to DEJ or skin. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width.

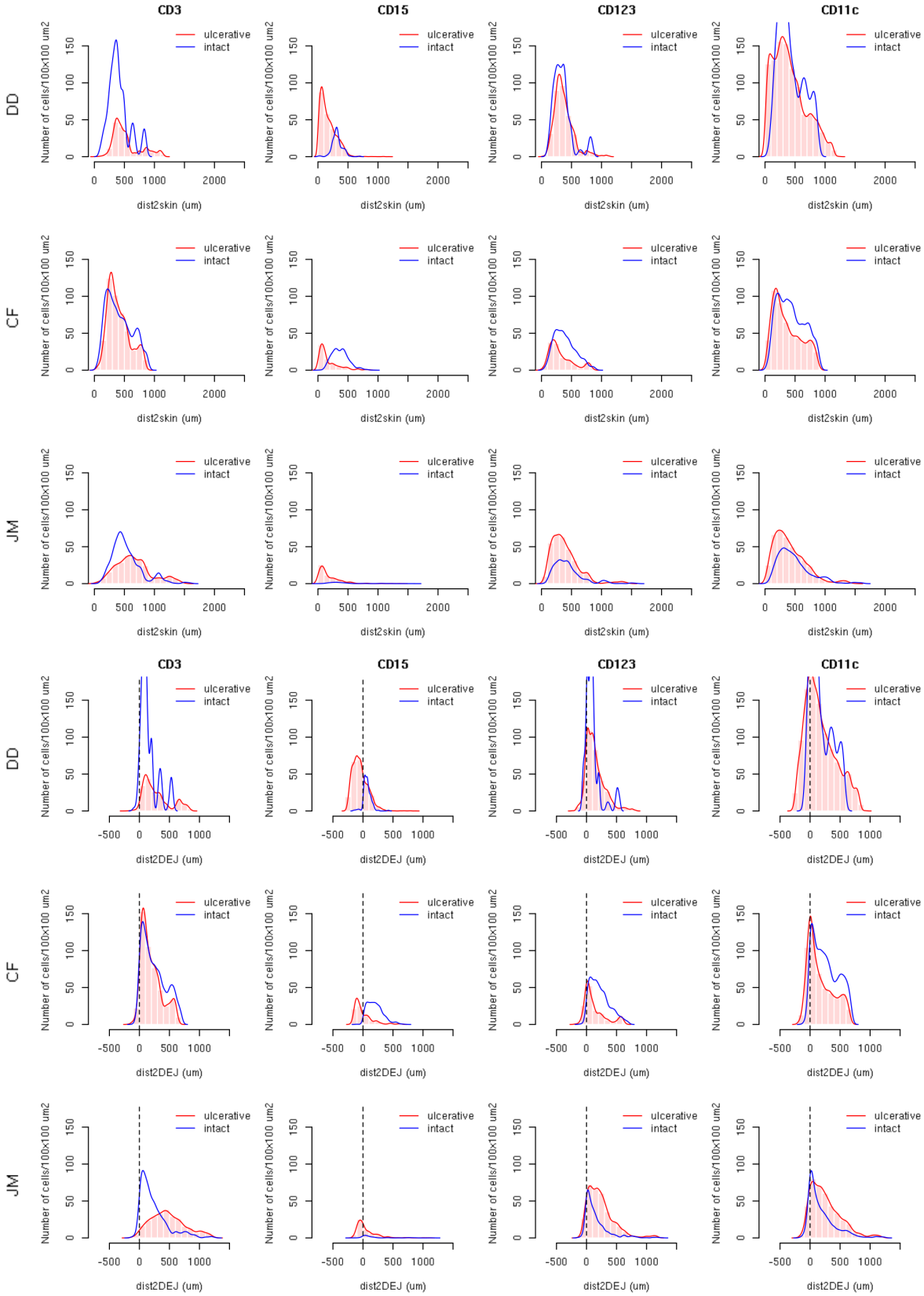



Figure 4: Number of cells per 100x100 um2 tissue as a function of distance to skin or DEJ. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width.

Comparing the first two rows in the first column, we see that the ratio of the area under the curve across both epi and dermis between red and blue(ulcerative and intact) is about 1 for CD3/CF and 1/3 for CD3/DD. This suggests that CD3 is depleted in lesion in DD, but not in CF.

Compare lesion with normal in CD15/CF, we see that the lesion peak is shifted to the left compared to the normal peak. This pattern is more clear in CF than in DD. We don't see this pattern for CD3.

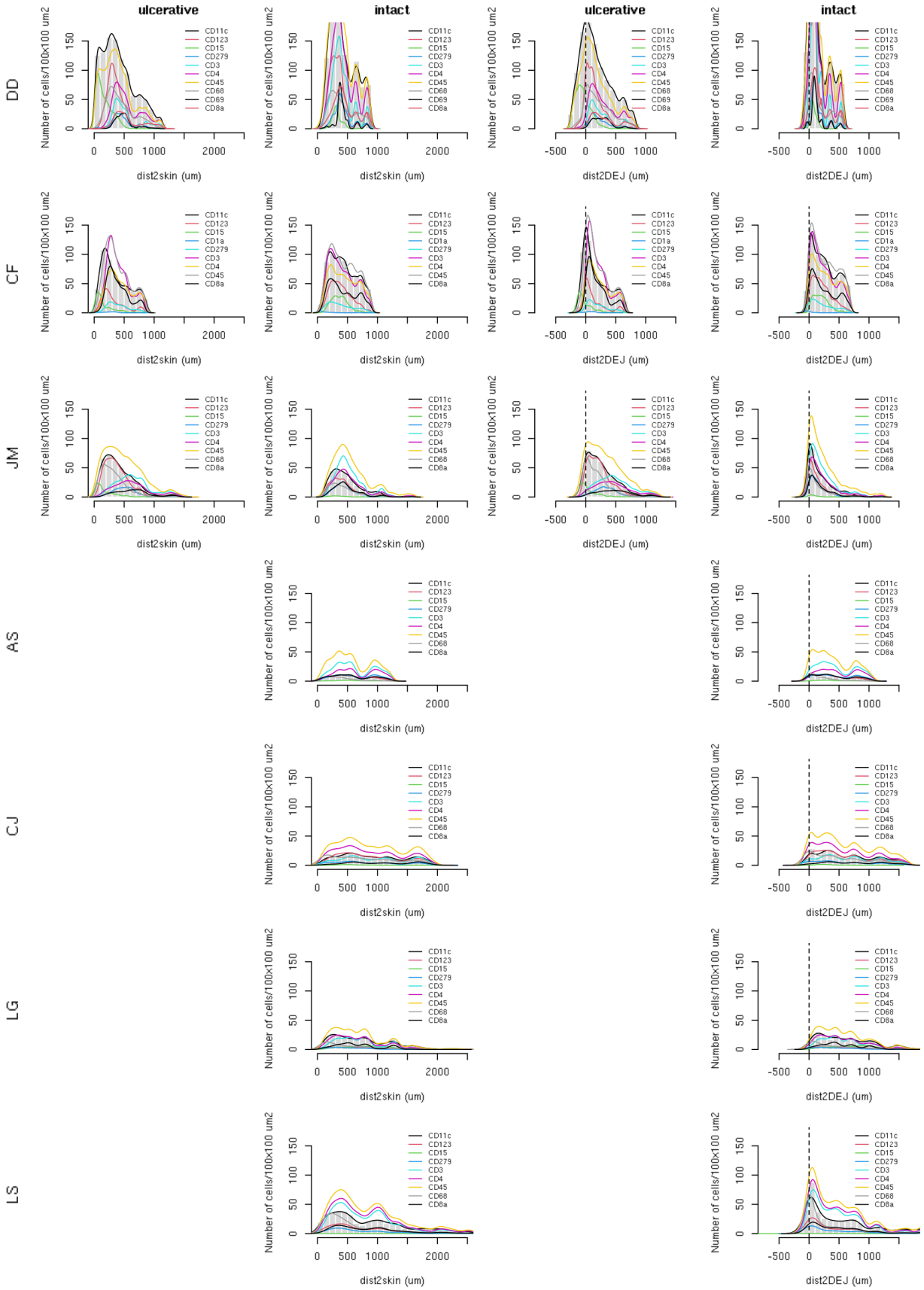
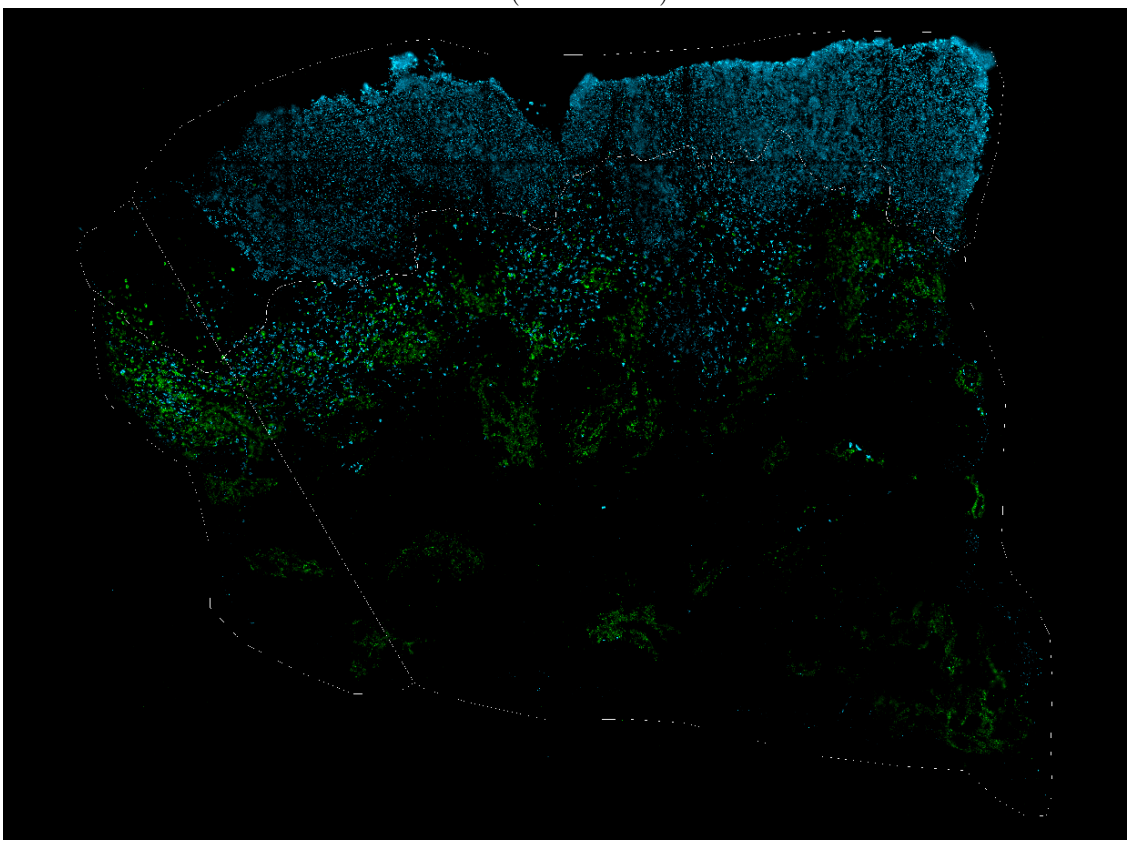



Figure 5: Number of cells per 100x100 um2 tissue as a function of distance to skin. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width.

That the CD15 dist2skin peak in DD/lesion is near skin while the corresponding dist2DEJ peak is near DEJ is due to the fact that the DEJ line is very curvy.

DD (6540 × 5262)



CF (5296 × 4040)

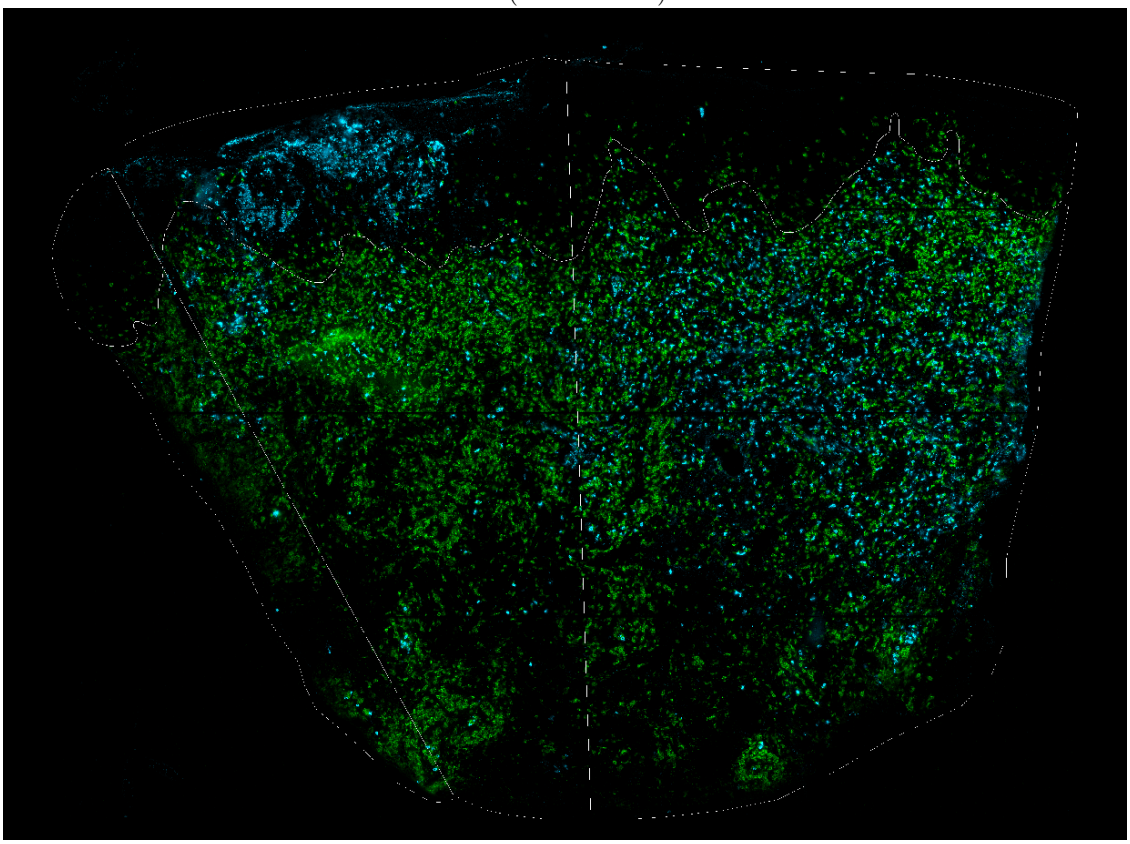


Figure 6: CD3: green, contrast limits 0-0.5; CD15: cyan. The intact/ulcerative boundary and the epidmeris/dermis boundary are shown in white. Outer boundaries are not tight.

JM (6528 × 4008)

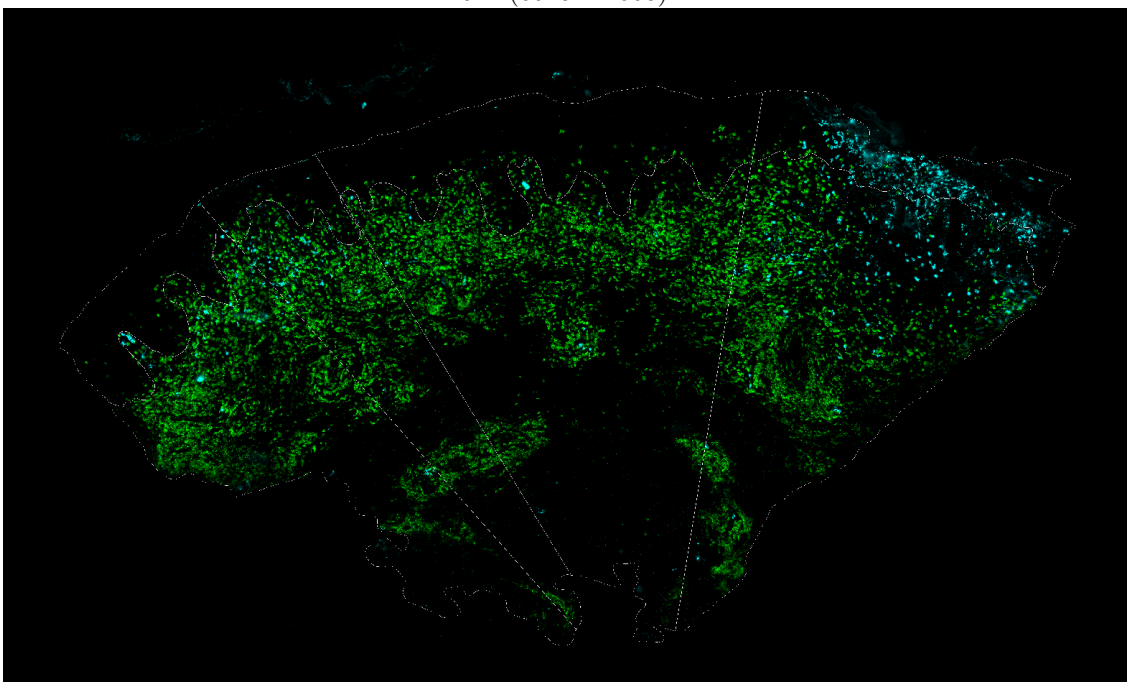


Figure 7: CD3: green; CD15: cyan, contrast limits 0-0.5. Region boudaries are shown in white.

In Figure 7 we see that DD lesion has less CD3 and is further away from DEJ than CF. Correspondingly, in Figure 1, we see that in the dermis region close to DEJ, the count density is about 50 for CF/ulcerative, CF/intact, and CF/ulcerative; but for DD/ulcerative, the count density is much lower.

Biospies with only intact regions seem to have lower number of immune cells.

# 4 Discussion

Neutraphil to lymphocyte ratio (NLR) in blood is a prognostic factor for cancer. Lower is better.

# 5 References

Chan A, Bentzen S, Rout A, Miller K. Predicting if lung cancer will relapse-the role of neutrophil/lymphocyte ratio. Hematol Oncol Stem Cell Ther. 2023;16(2):158–61.

Costa T, Nogueiro J, Ribeiro D, Viegas P, Santos-Sousa H. Impact of serum albumin concentration and neutrophil-lymphocyte ratio score on gastric cancer prognosis. Langenbecks Arch Surg. 2023;408(1):57.

Chen W, Xin S, Xu B. Value research of NLR, PLR, and RDW in prognostic assessment of patients with colorectal cancer. J Healthcare Eng. 2022;16(2022):7971415.